



S2.4 From models to materials (HL)

Structure 2 · Models of bonding and structure · IB Diploma · Concept guide and original exam-style practice

Understand the concept

The bonding models (ionic, covalent, metallic) can be extended to explain the structure and properties of real materials, including polymers (long chains formed by addition or condensation polymerisation, whose properties depend on chain length, branching and intermolecular forces between chains), alloys (mixtures of metals, or a metal with another element, where a different-sized atom distorts the regular metallic lattice and increases hardness), and nanoparticles (particles with at least one dimension below about 100 nm, which have a very high surface-area-to-volume ratio giving them different physical and chemical properties, including catalytic activity, compared with the bulk material). Choosing or engineering a material for a given application requires matching the bonding/structure to the required properties (e.g. strength, conductivity, flexibility, biodegradability).

Worked example: follow the reasoning

Explain why low-density poly(ethene) (LDPE), which has many branched chains, is more flexible and has a lower melting point than high-density poly(ethene) (HDPE), which has mostly linear, unbranched chains.

Step 1: Both are addition polymers of ethene, held together internally by strong covalent bonds along the carbon backbone.

Step 2: Between separate polymer chains, only weak London (dispersion) forces act.

Step 3: HDPE's linear chains can pack closely and align regularly, maximising the area of contact and so maximising the total London forces between chains.

Step 4: LDPE's branched chains cannot pack as closely or regularly, reducing the contact area and total intermolecular force between chains.

Answer: LDPE has weaker intermolecular forces between its branched chains than HDPE has between its closely packed linear chains, so LDPE is more flexible and softens/melts at a lower temperature.

Quick check · multiple choice

Which property change is typically observed when a bulk material is reduced to nanoparticle size?

- A Surface-area-to-volume ratio decreases
- B Surface-area-to-volume ratio increases, often increasing catalytic activity
- C Melting point always increases
- D Chemical reactivity always decreases

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

- 1 (a) Distinguish between addition and condensation polymerisation, giving the type of small molecule (monomer) needed for each. [3]
- (b) Nylon-6,6 is a condensation polymer with strong hydrogen bonding possible between adjacent chains, in addition to the covalent amide (peptide) linkages along each chain. Explain why nylon-6,6 has greater tensile strength than a simple addition polymer such as poly(ethene) of similar chain length. [3]
- (c) Suggest one advantage of using nanoparticle catalysts (e.g. nanoparticulate platinum) rather than the bulk metal in an industrial process. [1]

Final answer: _____



Answers and explanation

Question 1 [7]

(a) Addition polymerisation: monomers contain a C=C double bond that opens up and joins monomers together with no atoms lost, e.g. alkenes forming polyalkenes. Condensation polymerisation: monomers have two reactive functional groups (e.g. -OH, -COOH, or -NH₂) and join with the loss of a small molecule (commonly H₂O), e.g. diols/diacids or diamines/diacids forming polyesters/polyamides. [3]

(b) In nylon-6,6, the C=O and N-H groups along adjacent chains can form hydrogen bonds between chains, in addition to the covalent bonds within each chain; poly(ethene) chains have only weak London forces between chains. The additional, stronger intermolecular (hydrogen bonding) forces between nylon chains mean more energy/force is required to pull chains apart, giving nylon greater tensile strength. [3]

(c) Any valid answer, e.g.: nanoparticle catalysts have a much greater surface-area-to-volume ratio than the bulk metal, so a much smaller mass of (expensive) catalyst is needed to provide the same catalytic surface area/rate of reaction, reducing cost. [1]

Quick check answer

B. Reducing particle size greatly increases the surface-area-to-volume ratio, exposing proportionally more atoms at the surface, which commonly increases catalytic activity and reactivity.

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or content error, then retry the practice question without looking at the solution.

Based on the official IB Chemistry HL/SL subject guide.

Original FagusED teaching material. Not an official IB publication; question style informed by IB specimen papers but content and numbers are original.